Occlusion-Aware Frame Interpolation

A controlled study of lightweight learned blending

Which endpoint should contribute at each pixel?

Two video frames → frozen optical flow → shared warping → blending mask

+0.334 dB mean PSNR over fixed blending

3,782 test triplets · 95k trainable parameters · three mask seeds

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Related work: established principle, focused experiment

We claim neither a novel interpolation algorithm nor an exact reproduction of a prior study.

Prior work	Established idea	Our narrower scope
Super SloMo (2018)	Learned visibility + flow refinement	We freeze flow; train only the mask
Softmax Splatting (2020)	Weighted forward warping + synthesis	We use shared weighted splats
RIFE (2022)	Directly learned intermediate flow	We isolate blending with RAFT-small

Contribution: implementation, controlled comparisons, uncertainty and failure analysis.

References: Jiang et al., CVPR 2018 · Niklaus & Liu, CVPR 2020 · Huang et al., ECCV 2022
Paper links and implementation provenance are included in the notebook and README.

The three methods share the same motion and warps

Only the rule for choosing the blending mask changes in the primary comparison.

$$\hat{I}_t = M \odot W_0 + (1 - M) \odot W_1$$

Method	Mask	Learning
Fixed	M = 0.5 at every pixel	No training
Heuristic	Flow consistency + splat coverage	Strength selected on validation
Learned	18-channel input → small U-Net → sigmoid	95,185 trainable parameters

Frozen RAFT-small · half-flow forward splatting · maps aligned on the midpoint grid

A mask can select available information; it cannot reconstruct details missing from both warps.

Data integrity before model comparison

A corrected rerun of exploratory code, with a fixed and auditable protocol.

Split	Triplets	Use
Training	1,200	8k steps; 192×192 crops after full-frame warping
Validation	150	Checkpoint + heuristic selection; 64 for refiner tuning
Test	3,782	Full public Vimeo test mirror; never used for selection

- Pinned dataset revisions, sequence IDs and decoded-frame SHA-256
- Exact train/test overlap and repeated train/val frames excluded
- Three mask seeds; identical warps and metrics for every method

Directory-prefix separation is not proof of source-video independence. Similar scenes may remain.

A small learned mask improves the held-out reconstruction

Mean per-image PSNR/SSIM/MAE; regional PSNR uses pooled squared error.

Method	PSNR ↑	SSIM ↑	MAE ↓	PSNR dis ↑	PSNR occ ↑
Fixed	32.583	0.94730	0.01131	21.103	24.138
Heuristic	32.664	0.94769	0.01116	21.128	24.170
Learned (seed 42)	32.917	0.95076	0.01081	21.467	24.617

ΔPSNR vs fixed: +0.334 dB | paired 95% CI [+0.314, +0.356]

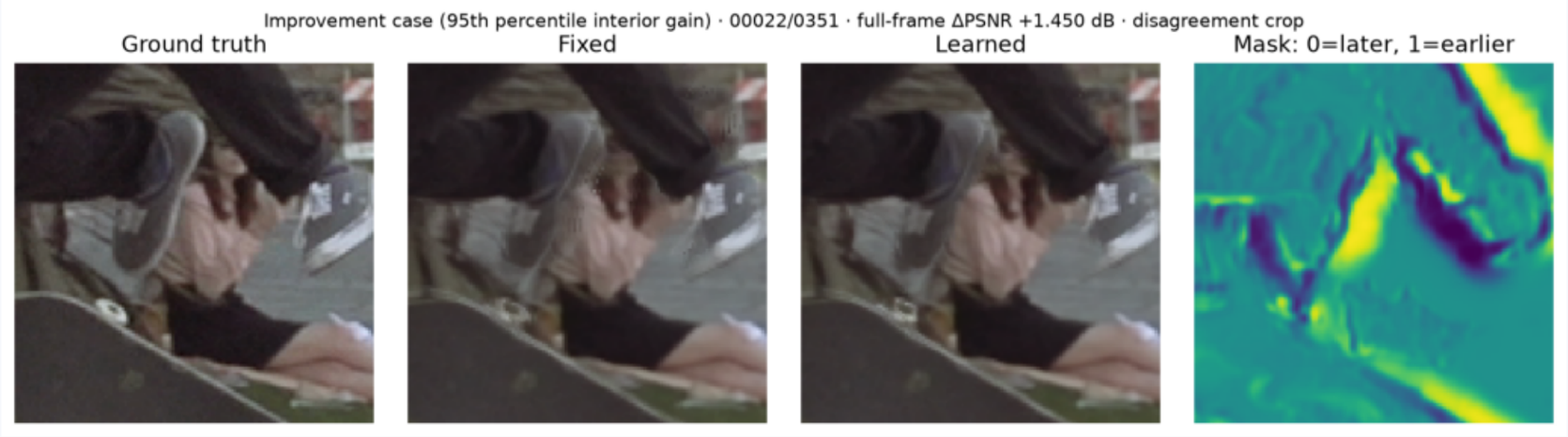
All three mask seeds: gains from +0.273 to +0.359 dB.

Post-hoc check without a 16px border: +0.234 dB, 95% CI [+0.215, +0.253].

dis / occ are disagreement and occlusion-related proxies, not ground-truth visibility labels.
Bootstrap resamples triplets and does not account for unknown video-level correlations.

Inspect both gains and failures

Post-hoc illustrations: 95th-percentile interior gain and 5th-percentile full-frame gain; labels show full-frame ΔPSNR.



Generative refinement is a separate extension

Same 3.31M-parameter refiner architecture; residual L1 regression vs conditional flow matching.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR dis ↑
Mask only	32.917	0.95077	0.0413	21.467
Mask + L1	33.405	0.95499	0.0342	21.926
Mask + FM	32.031	0.94003	0.0513	20.897
FM mean of 4	32.704	0.94775	0.0374	21.299
Validation-selected FM	32.031	0.94003	0.0513	20.897

Selected on validation: fm · euler 20 · guidance 1

FM mean-of-4 improves LPIPS over mask-only but loses PSNR; L1 is better on both.
Selected FM equals the baseline configuration. Fixed per-triplet noise; paired confidence intervals are in the notebook.

Diagnose the sampler before increasing training

Validation only: Euler/Heun × 20/50/100; CFG is evaluated only after condition-dropout training.

Solver	Steps	PSNR dis ↑	LPIPS window ↓	ms / triplet*
Euler	20	21.116	0.0727	55.9
Euler	50	20.953	0.0807	128.5
Euler	100	20.957	0.0811	244.9
Heun	20	20.939	0.0846	103.5
Heun	50	20.869	0.0862	242.4
Heun	100	20.915	0.0840	476.9

Additional ablations: EMA → logit-normal time → condition dropout / CFG; 12k vs 30k steps.

*Refiner-only batch-4 throughput measurement, not batch-1 end-to-end latency. Heun uses twice as many network calls per step.

All ablations and the exact validation selection are preserved in the notebook and CSV files.

Cost and the limits of a mask-only model

Runtime is measured on H100 at 256×448, batch size 1, with CUDA synchronization.

0.84 ms

mask + blending

56.04 ms

full two-frame interpolation

The learned mask closes 11.5% of the oracle MSE gap in disagreement regions.

- Highest-disagreement quartile: +0.209 dB; gains do not grow monotonically.
- The oracle bounds convex blending MSE and uses ground truth.
- Imperfect flow and missing detail limit a mask; FM adds repeated network calls.

Reported latency excludes video decoding/encoding. This prototype is not DLSS and makes no gaming real-time claim.

What this experiment establishes

A reproducible answer within a deliberately limited setup.

1 Learned blending improves reconstruction: +0.334 dB vs 50/50.

2 Gains are consistent across mask seeds, but failures and a large oracle gap remain.

3 Best tested refiner by mean test PSNR: l1.
Generative sampling must justify its quality and computational cost.

Limitations: small training subset · one refiner seed · proxy visibility regions · unknown scene correlations

Deliverables: executed notebook · source + checkpoints + per-triplet CSV · inference demo